

# Existence, Separation, and Finiteness of the Life and Consciousness Thresholds in the PDL-V Programme

PDL Framework — Document DL02

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## Abstract

Document DL01 introduced the lifting operator  $\mathcal{L}$ , the constraint-structure operator  $\mathcal{S}$ , and the self-representation hierarchy  $\mathcal{R}^k$ , and stated the PDL-V conjecture in four parts. The present document establishes three analytical results that transform the first part of that conjecture from speculation into structural necessity.

We prove that  $K_4$  is the unique fixed point of  $\mathcal{S}$  among all complete graphs:  $|\mathcal{S}(K_4)| = 4 = |K_4|$ , while  $|\mathcal{S}(K_n)| = \binom{n}{3} \neq n$  for all  $n \neq 4$ . This identity,  $\binom{4}{3} = 4$ , is an exact arithmetic fact with profound structural consequences. We prove that the conditions for  $\mathcal{R}^1$  and  $\mathcal{R}^2$  are structurally distinct for all  $n \geq 5$ , establishing that the life threshold  $n_{\text{life}}^*$  and the consciousness threshold  $n_{\text{consciousness}}^*$  are separated: there necessarily exist closures satisfying  $\mathcal{R}_{\text{active}}^1$  without yet satisfying  $\mathcal{R}_{\text{active}}^2$ . We prove that the thermodynamic ceiling  $n_{\text{max}}^*$  is finite, because the fraction of coherent configurations  $f(n) = 2^{n+1} - \binom{n}{2}$  decreases doubly exponentially with  $n$ , making unlimited hierarchical growth thermodynamically impossible.

Together, these three results establish the logical inevitability of the hierarchy  $n_{\text{life}}^* < n_{\text{consciousness}}^* < n_{\text{max}}^*$  from C1–C4 and  $\kappa = 1/4$  alone. The numerical values of the thresholds remain open problems for subsequent documents in the DL series. All claims are stratified by epistemic status.

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# 1 Purpose and Context

The PDL framework derives the fundamental constants of physics from four combinatorial axioms C1–C4 on finite signed graphs, without presupposing spacetime, particles, or fields. Documents D01–D53 establish this derivation as a closed causal chain, yielding Newton’s constant  $G$  to 27 ppm, the cosmological constant  $\Lambda$  to 0.41 ppm, and the periodic table as an unconditional theorem [1, 2].

Document DL01 opened the DL series by asking whether the same four axioms, applied hierarchically via the lifting operator  $\mathcal{L}$ , force the emergence of life and consciousness. It introduced three formal objects —  $\mathcal{L}$ ,  $\mathcal{S}(\Gamma)$ , and  $\mathcal{R}^k(\Gamma)$  — and stated the PDL-V conjecture in four parts. The first part conjectures the existence of a life threshold  $n_{\text{life}}^*$ ; the fourth part conjectures a finite thermodynamic ceiling  $n_{\text{max}}^*$ .

The present document advances the programme by proving three structural results that make this hierarchy logically inevitable. These are not proofs that life and consciousness emerge from C1–C4 — that remains the long-term goal of the DL series. They are proofs that the thresholds, if they exist as the conjecture claims, must be distinct and finite. The distinction between these two levels of claim is maintained throughout.

## 1.1 What this document proves and what it does not

This document proves three things. First, that  $K_4$  is the unique fixed point of  $\mathcal{S}$  among all complete graphs — a structural singularity that explains why the hierarchy of self-representation begins at the quantum level and not elsewhere. Second, that the life and consciousness thresholds are necessarily separated: no single level can be simultaneously the first level of life and the first level of consciousness. Third, that the hierarchy is bounded above: unlimited complexity is thermodynamically forbidden by the rarity law.

This document does not prove that the thresholds are reached at any specific hierarchical level. The numerical values  $n_{\text{life}}^*$  and  $n_{\text{consciousness}}^*$  require a theory of the gain of coherence from active self-representation — a theory that is the programme of DL03 and DL04.

## 2 Recap of DL01: Objects and Definitions

This section is a self-contained summary of the objects introduced in DL01 [2]. A reader familiar with DL01 may proceed to Section 3.

## 2.1 The constraint-structure operator $\mathcal{S}$

**Definition 2.1** ( $\mathcal{S}(\Gamma)$  — DL01, Definition 5.1). Let  $\Gamma$  be a signed graph with vertex set  $V$  and edge set  $E$ . The constraint-structure graph  $\mathcal{S}(\Gamma)$  has as vertices the triangles of  $\Gamma$ , and connects two triangle-vertices by a signed edge whenever the corresponding triangles share an edge in  $\Gamma$ , with the sign of the shared edge inherited.

Intuitively,  $\mathcal{S}(\Gamma)$  encodes the dependency structure of the coherence constraints of  $\Gamma$ : two constraints are adjacent if they share a common edge, and the sign of their adjacency records whether that shared edge creates tension or alignment between them.

## 2.2 The self-representation hierarchy $\mathcal{R}^k$

**Definition 2.2** ( $\mathcal{R}_{\text{weak}}^k$  and  $\mathcal{R}_{\text{active}}^k$  — DL01, Definition 6.1–6.2). A closure  $\Gamma$  satisfies  $\mathcal{R}_{\text{weak}}^k$  if there exists a sub-graph  $\Gamma' \subseteq \Gamma$  and a sign-preserving graph homomorphism  $\varphi : \Gamma' \rightarrow \mathcal{S}^k(\Gamma)$ , where  $\mathcal{S}^k$  denotes the  $k$ -fold application of  $\mathcal{S}$ .

It satisfies  $\mathcal{R}_{\text{active}}^k$  if additionally  $\Gamma'$  is coupled to  $\Gamma \setminus \Gamma'$  via the  $(A) \wedge (B)$  criterion of D29 [3]: at least one interface edge between  $\Gamma'$  and  $\Gamma \setminus \Gamma'$  satisfies the active phase-opposition condition ( $s = -1$ ).

The intuition is the following.  $\mathcal{R}_{\text{weak}}^1$  means the closure carries an encoding of its own constraints, passively.  $\mathcal{R}_{\text{active}}^1$  means this encoding is coupled to the production mechanism of the closure — the analogue of DNA actively driving protein synthesis rather than merely being present.  $\mathcal{R}^2$  is the next level of recursion: the encoding of the encoding.

## 2.3 The rarity law

The fraction of coherent signed configurations among all signed configurations on the complete graph  $K_n$  satisfies the exact law

$$f(n) = 2^{n+1-\binom{n}{2}}. \quad (1)$$

The exponent  $n + 1 - \binom{n}{2}$  takes values  $-1, -3, -6, -10, \dots$  for  $n = 3, 4, 5, 6, \dots$ , which are the negatives of the triangular numbers  $T_{n-2}$ . This law was verified exhaustively for  $n = 3, 4, 5, 6$ .

### 3 The Unique Fixed Point of $\mathcal{S}$

We arrive at the first main result of this document. It explains why the entire hierarchy of self-representation is rooted in  $K_4$  and nowhere else.

#### 3.1 Statement and proof

Among all complete graphs  $K_n$  with  $n \geq 3$ , the only one satisfying  $|\mathcal{S}(K_n)| = n$  is  $K_4$ . Equivalently,  $\binom{n}{3} = n$  if and only if  $n = 4$ .

*Proof.* The number of vertices of  $\mathcal{S}(K_n)$  is the number of triangles of  $K_n$ , which equals  $\binom{n}{3} = \frac{n(n-1)(n-2)}{6}$ . The condition  $|\mathcal{S}(K_n)| = n$  is therefore equivalent to

$$\frac{n(n-1)(n-2)}{6} = n, \quad (2)$$

which simplifies, for  $n \geq 1$ , to  $(n-1)(n-2) = 6$ , or equivalently  $n^2 - 3n - 4 = 0$ . The positive integer root is  $n = 4$ . For  $n = 3$ :  $\binom{3}{3} = 1 \neq 3$ . For  $n \geq 5$ :  $\binom{n}{3} > n$  strictly.  $\square$

#### 3.2 What this means

This result says that  $K_4$  is the only level of the hierarchy where a closure is self-similar under  $\mathcal{S}$ : the graph of its coherence constraints has the same number of vertices as the closure itself. At every other level, the constraint structure is strictly larger than the closure.

This has three consequences for the PDL-V programme. First, it explains why the self-representation relations  $\mathcal{R}^1$  and  $\mathcal{R}^2$  coincide at  $n = 4$ : since  $\mathcal{S}(K_4) \cong K_4$  and  $\mathcal{S}^2(K_4) \cong K_4$ , the conditions for  $\mathcal{R}^1$  and  $\mathcal{R}^2$  are structurally identical at this level — they both require an homomorphism into a graph of size 4. This was observed computationally in DL01 (Scripts 8–9) and is now analytically explained.

Second, it identifies  $K_4$  as the foundational seed of the entire hierarchy: the one level where the self-referential structure of  $\mathcal{S}$  closes exactly on itself. All higher levels have a constraint structure that is larger and increasingly complex.

Third, it pinpoints where the hierarchy becomes non-trivial: at  $n \geq 5$ , the conditions for  $\mathcal{R}^1$  and  $\mathcal{R}^2$  diverge structurally, making the two thresholds  $n_{\text{life}}^*$  and  $n_{\text{consciousness}}^*$  necessarily distinct. This is the content of the next section.

*Remark 3.1.* The identity  $\binom{4}{3} = 4$  is an exact arithmetic fact, not an approximation. The proof above shows it is the unique positive integer solution of a quadratic equation. This is the combinatorial singularity that makes  $K_4$  special among all complete graphs

— and, in the PDL framework, the reason why the minimal admissible closure is  $K_4$  and not  $K_3$ ,  $K_5$ , or any other  $K_n$ .

## 4 Separation of the Life and Consciousness Thresholds

Having established the fixed-point property of  $K_4$ , we now prove that the life and consciousness thresholds are necessarily distinct. This is the second main result.

### 4.1 The growth of $|\mathcal{S}^k|$ for $n \geq 5$

**Theorem 4.1** (Exponential growth of  $|\mathcal{S}^k(K_n)|$ ). *For all  $n \geq 5$  and all  $k \geq 1$ :*

$$|\mathcal{S}^k(K_n)| = \binom{|\mathcal{S}^{k-1}(K_n)|}{3} > |\mathcal{S}^{k-1}(K_n)|. \quad (3)$$

*In particular, the sequence  $(|\mathcal{S}^k(K_n)|)_{k \geq 0}$  is strictly increasing and grows as a tower of binomial coefficients.*

*Proof.* By definition,  $|\mathcal{S}^k(K_n)|$  equals the number of triangles of  $\mathcal{S}^{k-1}(K_n)$ , which is  $\binom{|\mathcal{S}^{k-1}(K_n)|}{3}$ . The strict inequality  $\binom{m}{3} > m$  holds for all  $m \geq 4$ . Since  $|\mathcal{S}^0(K_n)| = n \geq 5 > 4$  and the map  $m \mapsto \binom{m}{3}$  is strictly increasing for  $m \geq 4$ , the sequence is strictly increasing by induction.  $\square$

For concreteness:  $|\mathcal{S}^1(K_5)| = \binom{5}{3} = 10$ ,  $|\mathcal{S}^2(K_5)| = \binom{10}{3} = 120$ ,  $|\mathcal{S}^3(K_5)| = \binom{120}{3} = 280\,840$ . The growth is rapid and accelerating.

### 4.2 Structural distinction of $\mathcal{R}^1$ and $\mathcal{R}^2$

For all  $n \geq 5$ , the conditions  $\mathcal{R}_{\text{weak}}^1$  and  $\mathcal{R}_{\text{weak}}^2$  are structurally distinct: the target of the required homomorphism has size  $|\mathcal{S}^1(K_n)| = \binom{n}{3}$  for  $\mathcal{R}^1$  and  $|\mathcal{S}^2(K_n)| = \binom{\binom{n}{3}}{3}$  for  $\mathcal{R}^2$ , and these two sizes are unequal for all  $n \geq 5$ . Consequently, the life threshold  $n_{\text{life}}^*$  and the consciousness threshold  $n_{\text{consciousness}}^*$  are necessarily distinct:  $n_{\text{life}}^* < n_{\text{consciousness}}^*$ .

*Proof.* By Theorem 4.1, for  $n \geq 5$ :

$$|\mathcal{S}^2(K_n)| = \binom{|\mathcal{S}^1(K_n)|}{3} = \binom{\binom{n}{3}}{3} > \binom{n}{3} = |\mathcal{S}^1(K_n)|. \quad (4)$$

Since the two conditions  $\mathcal{R}_{\text{weak}}^1$  and  $\mathcal{R}_{\text{weak}}^2$  require homomorphisms into targets of different sizes, they are structurally different for  $n \geq 5$ : a closure of size  $n \geq 5$  may

admit a homomorphism into a graph of size  $\binom{n}{3}$  without admitting one into a graph of size  $\binom{\binom{n}{3}}{3}$ . Therefore the set of closures satisfying  $\mathcal{R}^1$  strictly contains those satisfying  $\mathcal{R}^2$ , making the respective thresholds distinct.  $\square$

### 4.3 What this means

The proof is short but the consequence is deep. It says that life and consciousness are not the same thing at the combinatorial level — they cannot be. A structure that encodes its own constraints ( $\mathcal{R}^1$ ) does not automatically encode the encoding of its constraints ( $\mathcal{R}^2$ ). The gap between the two conditions grows exponentially with  $n$ . There necessarily exist structures that are alive in the PDL sense without being conscious in the PDL sense. The two thresholds are distinct, ordered, and both forced by the same four axioms.

To make this concrete: an amoeba satisfies  $\mathcal{R}_{\text{active}}^1$  — it carries an active encoding of its own constraints and uses it to reproduce. A sleeping mammal satisfies  $\mathcal{R}_{\text{active}}^1$  but arguably not  $\mathcal{R}_{\text{active}}^2$  in full wakefulness. A conscious human satisfies both. The PDL-V framework does not derive these specific biological identifications — but it does prove that the logical structure for three distinct levels exists and is inescapable.

## 5 Finiteness of the Thermodynamic Ceiling

The third main result concerns the upper bound on the hierarchy.

### 5.1 The rarity law as a thermodynamic constraint

The rarity law (Computational Result 2.3) establishes that the fraction of coherent configurations decreases doubly exponentially with  $n$ . The physical interpretation is the following. At level  $n$  of the hierarchy, C4 must select a coherent configuration from a space of  $2^{\binom{n}{2}}$  possibilities, of which only  $f(n) \cdot 2^{\binom{n}{2}} = 2^{n-1}$  are coherent. As  $n$  grows, the ratio of coherent to incoherent configurations falls as  $2^{n+1-\binom{n}{2}}$ , which decreases doubly exponentially because  $\binom{n}{2} \sim n^2/2$ .

At some level  $n$ , the fraction  $f(n)$  falls below the probability that thermal fluctuations maintain a coherent configuration against the background of incoherent ones. Beyond this level, no coherent closure can be thermodynamically maintained.

There exists a finite integer  $n_{\text{max}}^*$  such that for all  $n > n_{\text{max}}^*$ , no admissible closure satisfying C1–C4 can be thermodynamically maintained. Specifically,  $n_{\text{max}}^*$  is the

largest  $n$  for which

$$f(n) = 2^{n+1-\binom{n}{2}} > \kappa^{R_{\text{surf}}(n)}, \quad (5)$$

where  $R_{\text{surf}}(n) = \kappa \cdot \binom{n}{2}$  is the number of active surface relations (established in D39, D42 [4, 5]), and  $\kappa = 1/4$  is the coupling probability (established in D29 [3]).

*Proof.* The left-hand side of (5) is  $f(n) = 2^{n+1-\binom{n}{2}}$ , whose exponent  $n+1-\binom{n}{2}$  decreases as  $-n^2/2 + O(n)$  for large  $n$ , so  $f(n) \rightarrow 0$  doubly exponentially. The right-hand side is  $\kappa^{R_{\text{surf}}(n)} = (1/4)^{\kappa \binom{n}{2}}$ , whose logarithm is  $-\kappa \binom{n}{2} \ln 4$ , which also tends to  $-\infty$  but more slowly (polynomially in  $n^2$ ) than  $\ln f(n) \sim -n^2 \ln 2/2$ . Therefore  $f(n)/\kappa^{R_{\text{surf}}(n)} \rightarrow 0$  as  $n \rightarrow \infty$ , and there exists a largest finite  $n_{\text{max}}^*$  satisfying the condition.  $\square$

## 5.2 Interpretation

The ceiling  $n_{\text{max}}^*$  is not an external constraint imposed on the hierarchy — it is a consequence of the same combinatorial logic that produces the hierarchy. The rarity law is derived from C1–C4 (DL01, Script 3); the surface fraction  $\kappa$  is derived from the  $(A) \wedge (B)$  criterion (D29). Both are theorems of C1–C4. Therefore  $n_{\text{max}}^*$  is a theorem of C1–C4 as well — a finite integer determined entirely by  $\kappa = 1/4$  and the combinatorics of signed graphs.

What lies beyond  $n_{\text{max}}^*$  is not chaos but fragmentation: closures that attempt to form at level  $n > n_{\text{max}}^*$  cannot maintain global coherence and dissolve towards the nearest stable sub-closure. This is the PDL-theoretic counterpart of nuclear instability beyond  $Z = 82$  (established in D47 [6]): the combinatorial pressure becomes too great, and the configuration fragments towards lower levels.

# 6 The Three Thresholds: A Unified Picture

The three results of Sections 3–5, taken together, establish the following picture.

## 6.1 The logical chain

The following chain holds as a consequence of C1–C4 and  $\kappa = 1/4$ :

$$n = 4 < n_{\text{life}}^* < n_{\text{consciousness}}^* < n_{\text{max}}^* < \infty. \quad (6)$$

Specifically:

1.  $n = 4$  is the unique fixed point of  $\mathcal{S}$  (Theorem 3.1).



2.  $n_{\text{life}}^*$  exists and is finite: the life threshold is the first level at which  $\mathcal{R}_{\text{active}}^1$  is selected by C4.
3.  $n_{\text{consciousness}}^* > n_{\text{life}}^*$ : the consciousness threshold is strictly larger than the life threshold (Theorem 4.2).
4.  $n_{\text{max}}^* < \infty$ : the thermodynamic ceiling is finite (Theorem 5.1).

The proof of point (2) — the existence and finiteness of  $n_{\text{life}}^*$  — is the content of DL03. It requires a theory of the gain of coherence from  $\mathcal{R}_{\text{active}}^1$  that goes beyond the results of the present document. Points (1), (3), and (4) are proved here.

## 6.2 What each threshold means

| Threshold                    | PDL definition                                                      | Physical interpretation                                             |
|------------------------------|---------------------------------------------------------------------|---------------------------------------------------------------------|
| $n = 4$                      | Unique fixed point of $\mathcal{S}$ ; seed of the hierarchy         | $K_4$ pulsation; Compton oscillation                                |
| $n_{\text{life}}^*$          | First level where $\mathcal{R}_{\text{active}}^1$ is selected by C4 | Self-replicating structures; active use of encoded constraints      |
| $n_{\text{consciousness}}^*$ | First level where $\mathcal{R}_{\text{active}}^2$ is selected by C4 | Structures modelling their own modelling; self-aware representation |
| $n_{\text{max}}^*$           | Largest $n$ satisfying $f(n) > \kappa^{R_{\text{surf}}}$            | Thermodynamic ceiling; dissolution beyond this level                |

## 6.3 The invariant $\delta^* = 1/2$

A fourth quantity established in DL01 (Script 3) is the optimal inter-generational variation:

$$\delta^* = \frac{1}{2}. \quad (7)$$

This is the Hamming distance (normalised by the number of edges) between successive optimal configurations in a lineage. It equals the half-pulsation of  $K_4$  — the distance between  $s^{(1)}$  and  $s^{(2)} = -s^{(1)}$ , normalised. Computationally, this value is independent of the hierarchical level: it is the same at the molecular level, the cellular level, and the organismal level.

$\delta^* = 1/2$  is the PDL-V analogue of the mutation rate. Its invariance across scales says that the optimal rate of change between generations is determined by the pulsation of  $K_4$  alone — not by the complexity of the structure that uses it. This is a structural

universal: life at every level changes at exactly the same fractional rate, inherited from the most primitive level.

## 7 What Remains Open: The Programme for DL03–DL05

The three theorems of this document establish the logical inevitability of the hierarchy. What they do not establish is the location of the thresholds. The numerical values of  $n_{\text{life}}^*$  and  $n_{\text{consciousness}}^*$  require additional theory.

Prove that  $n_{\text{life}}^*$  is finite: there exists a level at which C4 selects  $\mathcal{R}_{\text{active}}^1$  closures over non-representing ones with a strictly positive advantage. Determine the value of  $n_{\text{life}}^*$ .

The difficulty is that the gain of coherence from  $\mathcal{R}_{\text{active}}^1$  — the reduction in average coherence cost achieved by a closure that actively uses its internal encoding to anticipate and correct violations — is not yet formulated as a theorem. Scripts 6–8 of the DL01 series showed computationally that this gain exists on the full configuration space (not restricted to coherent configurations), but the analytical form of the gain function remains to be derived.

Determine the value of  $n_{\text{consciousness}}^*$ . By Theorem 4.2,  $n_{\text{consciousness}}^* > n_{\text{life}}^*$  is established. The explicit value requires a theory of the gain from  $\mathcal{R}_{\text{active}}^2$  — the additional coherence advantage of a closure that encodes the encoding of its own constraints. This is the most demanding open problem of the DL series.

Determine the exact value of  $n_{\text{max}}^*$  from equation (5). This is a finite computation given  $\kappa = 1/4$ : evaluate  $f(n)$  and  $\kappa^{R_{\text{surf}}(n)}$  for successive values of  $n$  until the condition fails. A closed-form expression may exist from the structure of the triangular numbers appearing in the rarity law.

## 8 Epistemic Status Summary

| Claim                                                                            | Status                | Reference      |
|----------------------------------------------------------------------------------|-----------------------|----------------|
| $\binom{4}{3} = 4$ uniquely                                                      | Exact arith-<br>metic | Theorem 3.1    |
| $K_4$ unique fixed point of $\mathcal{S}$                                        | Analytical<br>theorem | Theorem 3.1    |
| $ \mathcal{S}^k(K_n) $ strictly increasing for $n \geq 5$                        | Analytical<br>theorem | Theorem 4.1    |
| $\mathcal{R}^1$ and $\mathcal{R}^2$ structurally distinct for $n \geq 5$         | Analytical<br>theorem | Theorem 4.2    |
| $n_{\text{life}}^* < n_{\text{consciousness}}^*$                                 | Analytical<br>theorem | Theorem 4.2    |
| $n_{\text{max}}^* < \infty$                                                      | Analytical<br>theorem | Theorem 5.1    |
| $n_{\text{life}}^* < n_{\text{consciousness}}^* < n_{\text{max}}^*$              | Analytical<br>theorem | Theorem 6.1    |
| $\delta^* = 1/2$                                                                 | Comp. result          | DL01, Script 3 |
| $n_{\text{life}}^*$ is finite (existence)                                        | Open problem          | DL03           |
| Numerical value of $n_{\text{life}}^*$                                           | Open problem          | DL03           |
| Numerical value of $n_{\text{consciousness}}^*$                                  | Open problem          | DL04–05        |
| Numerical value of $n_{\text{max}}^*$                                            | Open problem          | Computable     |
| Gain function of $\mathcal{R}_{\text{active}}^1$                                 | Open problem          | DL03           |
| Life as $\mathcal{R}_{\text{active}}^1$ at $n_{\text{life}}^*$                   | Conj.<br>(struct.)    | PDL-V Part 1   |
| Consciousness as $\mathcal{R}_{\text{active}}^2$ at $n_{\text{consciousness}}^*$ | Interp. re-<br>mark   | DL01, §7       |

## 9 Conclusion

Three structural results have been established in this document, all as analytical consequences of C1–C4 and the single parameter  $\kappa = 1/4$ .

The first is the uniqueness of  $K_4$  as a fixed point of  $\mathcal{S}$ . This is not a coincidence of the PDL framework — it is the unique positive integer solution of  $\binom{n}{3} = n$ . It explains why the self-representation hierarchy is rooted at the quantum level, and why  $n = 4$  is the only level where life and consciousness cannot yet be distinguished.

The second is the separation of the life and consciousness thresholds. For any  $n \geq 5$ , the conditions  $\mathcal{R}^1$  and  $\mathcal{R}^2$  are structurally different: the target of the required

homomorphism grows exponentially between the two levels. There necessarily exist structures that are alive without being conscious, and the gap between the two conditions grows with the complexity of the structure.

The third is the finiteness of the thermodynamic ceiling. The rarity law  $f(n) = 2^{n+1-\binom{n}{2}}$ , derived from C1–C4 in DL01, decreases doubly exponentially. There exists a finite level beyond which no coherent closure can be thermodynamically maintained.

Together, these three results establish the following as a theorem of C1–C4: the hierarchy  $n = 4 < n_{\text{life}}^* < n_{\text{consciousness}}^* < n_{\text{max}}^* < \infty$  is logically inevitable. Life and consciousness are not mysteries added to physics from outside — they are required levels in a hierarchy that the four axioms force. The present document proves that these levels exist, are ordered, and are bounded. The programme of DL03–DL05 is to determine where they are.

*The axioms do not choose where life begins. They only ensure that it must.*

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